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Every camera is a structural health sensor.

Phase-based motion magnification turns the phones and CCTV cities already own into non-contact vibration sensors — with a damage alert, not just a spectrum.

AI ARENA 3.0 · PHYSICAL PROTOTYPE · AI VISION

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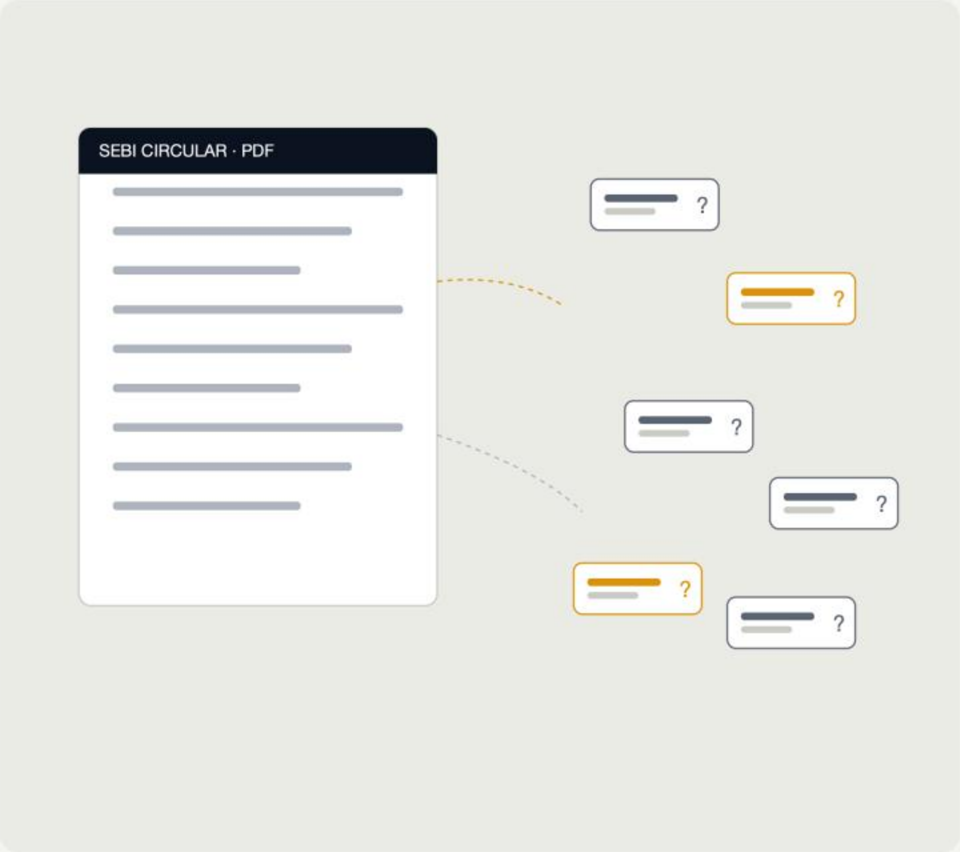
01 — PROBLEM UNDERSTANDING

India inspects its bridges by eye.

Morbi 2022 — 135 dead. Gambhira 2025 — 22 dead. After Gambhira, Gujarat audited 1,800+ bridges in panic: 20 closed outright, 113 restricted.

Contact monitoring costs lakhs per span, so the highway network gets sensors — municipal bridges get eyes only.



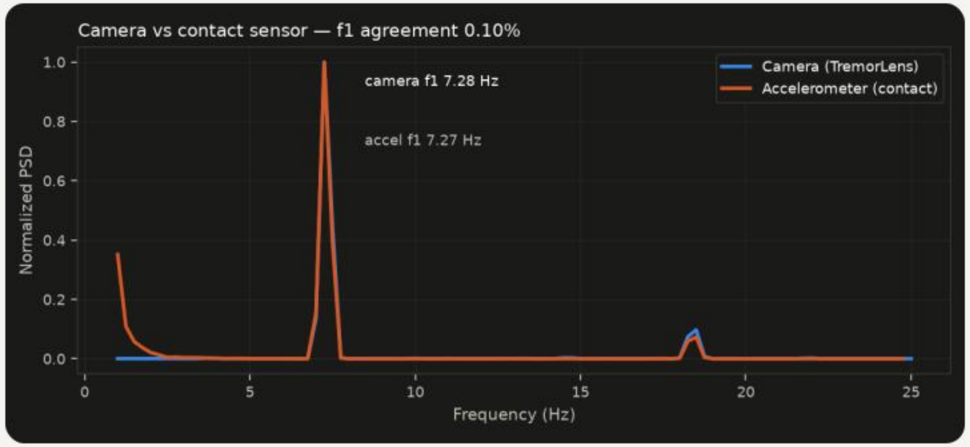
02 — **CONCEPT DEVELOPMENT**

Structures have a heartbeat. Cameras can hear it.

Every structure vibrates at natural frequencies set by its stiffness. Damage changes stiffness — the heartbeat shifts.

TremorLens reads that heartbeat from sub-pixel motion in ordinary video (MIT lineage: Wu 2012, Wadhwa 2013), scores drift against a healthy baseline, and raises the alert itself.

Chosen over 19 scored alternatives — the full decision board ships in our documentation.



Camera vs contact accelerometer — same heartbeat, 0.10% apart.

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03 — SYSTEM DESIGN — HIGH-LEVEL

One pipeline, camera to verdict.

Invisible to the eye. Unmistakable in the spectrum.

Watch — any 60 fps camera plus a ₹5 speckle sticker.

Measure — 1/50th-pixel motion, shake-cancelled by static references.

Fingerprint — the modal spectrum, the heartbeat.

Judge — SHS vs the stored healthy baseline.

03 — SYSTEM DESIGN — LOW-LEVEL

Built honest at every layer.

01
The rig
Twin 60 cm model trusses, DC vibration motor + tap, phone accelerometer (phyphox) as ground truth, DC lighting, tripod.

02
The vision
Auto-ROI on speckle targets, 50× upsampled phase correlation, static-reference cancellation of camera shake, Welch PSD.

03
The engine
Structural Heartbeat Score: 100 – 5× drift beyond a calibrated noise floor. HEALTHY / WATCH / ALERT. Baseline persisted as JSON.

04
Safety & IP
No people in frame, all processing local (DPDP-safe). Open MIT-lineage methods, cited; no vendor trademarks or workflow cloning.

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04 — MVP — MINIMUM VIABLE PROTOTYPE

Hands-free, launch to alert.

Zero keystrokes after launch: auto-ROI, auto-baseline, live monitor, automatic alert.

The command centre supervises the whole fleet on existing CCTV — one operator, every bridge.

Live demo runs from webcam or recorded rig footage — same code path.

05 — TESTING & VALIDATION

Validated against ground truth.

01
0.31%
frequency error vs known truth — stress-tested at 3× noise, 4× flicker, 4× camera shake: unchanged.

02
0.10%
camera vs contact accelerometer agreement (phyphox on the deck, displacement-equivalent spectra).

03
SHS 28
damaged twin flagged ALERT at −14.5% drift; healthy re-take stays HEALTHY at SHS 100 — no false positive.

04
< 1.2%
worst case across the full matrix (tap excitation, fewer cycles). Every run, every number, regenerates from one command.

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05 — TESTING — THE BLIND REVEAL

Three bridges. One secret fault.

One of three was tampered — the pipeline was not told which.

Bridge A	7.277 Hz	SHS 100	HEALTHY
Bridge B	7.277 Hz	SHS 100	HEALTHY
Bridge C	6.222 Hz	SHS 28	ALERT

It picked Bridge C.
It was Bridge C.

The tamper is structurally significant by design — the literature (and our docs) say single-bolt looseness is not honestly detectable. We claim exactly what we can prove.

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06 — ITERATION & EXECUTION

Broken three times. Fixed with physics.

01

Lattice hop

Checkerboard targets broke sub-pixel tracking — periodic patterns alias. Switched to aperiodic DIC speckle; now a setup rule.

02

Honest pixels

Interpolated test motion blurs instead of shifting. We supersample 8× and integer-shift — test data moves like real optics.

03

The f^4 trap

Raw accel spectra mislead: acceleration □ f^2 × motion, higher modes dominate. Fusion converts to displacement-equivalent.

04

Discipline

Verification harness ran before every commit. Every claim on every slide regenerates from code in the submission zip.

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07 — WHY IT MATTERS AT SCALE

The screening layer India never had.

01

1,72,517

national-highway structures inventoried — contact-sensor monitoring reaches only the newest spans.

02

₹0

marginal hardware where CCTV already exists. Vision instrumentation measured ~17× cheaper than contact rigs (peer-reviewed).

03

8–9 mo

Seoul ran continuous CCTV displacement monitoring on live bridges — precedent that city cameras can do structural duty.

04

Then escalate

TremorLens screens the fleet; only red-flagged spans get full contact instrumentation. Same engine for machines.

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08 — WHY THIS SCORES

Mapped to your rubric, deliberately.

01

Novelty

No product ships modal-fingerprint alerts on commodity cameras — the industrial tool costs $32k; we cite the MIT lineage.

02

Usability

Runs smooth without user intervention — filmed with zero keystrokes after launch. Baseline, monitor, alert: all automatic.

03

Innovation

Bridges, machines, towers, cable forces, city fleets — one engine, many real problems, at ₹0 marginal hardware.

04

Documentation

Docs follow your categories verbatim; negative results included; full repro steps; source code in the zip.

Roadmap: laser SpeckleBoost (nanometre mode) · moiré strain stickers · acoustic tap mode · acoustic camera fleet.

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Thank you.

TremorLens — hear the heartbeat of infrastructure, before it stops.

Working prototype · verified pipeline · command centre · full docs
github.com/mightbeanshuu/tremorlens (source in submission zip)

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